

Seminar 9

LR(0)

PRINCIPLE

- current state
- current symbol
- prediction that uniquely determines $\begin{cases} \nearrow \text{action to} \\ \searrow \text{move to another} \\ \text{state} \end{cases}$

LR(0) steps

1. Define items
2. Construct set of states
3. Construct table
4. Parse sequences based on moves between config.

Example:

$$S' \rightarrow S$$

$$S \rightarrow aA \quad (1)$$

$$A \rightarrow bA \quad (2)$$

$$A \rightarrow c \quad (3)$$

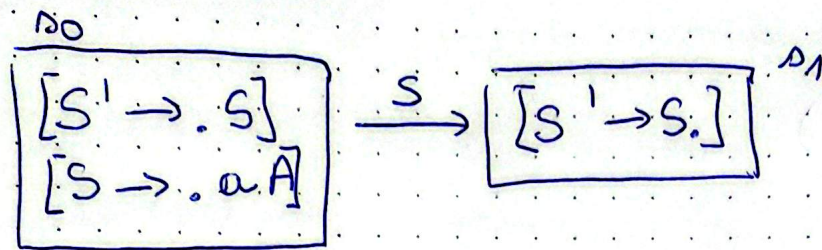
Items and states

$$s_0 = \text{closure}(\{[S' \rightarrow \cdot S]\}) = \{[S' \rightarrow \cdot S], [S \rightarrow \cdot aA]\}$$

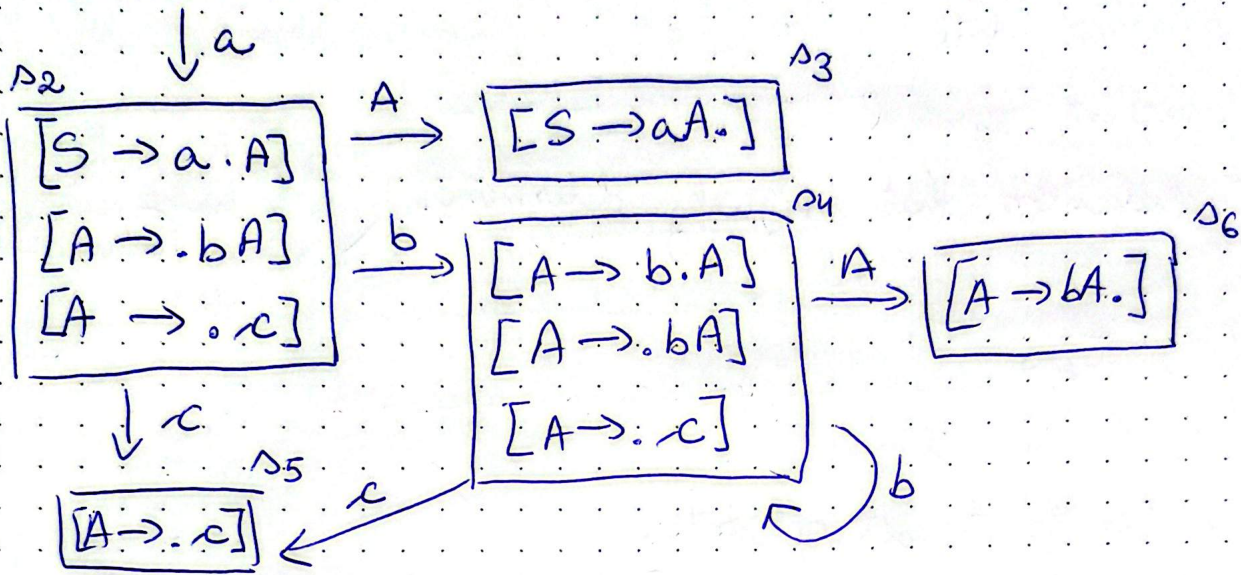
$$s_1 = \text{goto}(s_0, S) = \text{closure}([S' \rightarrow S \cdot]) = \{[S' \rightarrow S \cdot]\}$$

$$\begin{aligned} s_2 &= \text{goto}(s_0, a) = \text{closure}([S \rightarrow a \cdot A]) = \\ &= \{[S \rightarrow a \cdot A], [A \rightarrow \cdot bA], [A \rightarrow \cdot c]\} \end{aligned}$$

and so on...



Collection of canonical states



LR(0) TABLE

Actions

- shift : if $[A \rightarrow \alpha \cdot \beta] \in S_i$
- reduce if $[A \rightarrow \beta \cdot] \in S_i$ & $A \neq S'$
- accept if $[S' \rightarrow S \cdot] \in S_i$
- error otherwise

STATE	ACTION	GOTO				
		a	b	c	S	A
0	shift	2			1	
1	acc					
2	shift		4	5		3
3	reduce 1					
4	shift		4	5		6
5	reduce 3					
6	reduce 2					

PARSING

$w = abbc$

α : working stack β input stack

initial config: $(\$s_0, w\$, \epsilon)$

final config: $(\$s_{acc}, \$, \pi)$ output (production number sequence)

α	β	π
$\$0$	<u>a</u> bb c \$	ϵ
$\$0a2$	<u>b</u> bc\$	ϵ
$\$0a2b4$	<u>b</u> c\$	ϵ
$\$0a2b4b4$	<u>c</u> \$	ϵ
$\$0a2b4b4c5^{(1)}$	\$	ϵ
$\$0a2b4b4A6^{(2)}$	\$	3
$\$0a2b4A6$	\$	2, 3
$\$0a2A3$	\$	2, 2, 3
$\$0s1$	\$	1, 2, 2, 3
$\$acc$	\$	

(1) reduce with production 3 $\Rightarrow$ replace $c5$ with left hand side of production 3 which is A

After A write 6 because $s4$ with A will shift to $s6$

$\$0a2b4b4\cancel{c5} \rightarrow \$0a2b4b4A6$
 $\quad \quad \quad A \quad \quad \quad \downarrow \nearrow$
 $\quad \quad \quad \quad \quad \quad \quad 6$

(2) reduce with production 2 ($A \rightarrow bA$)

so replace $b4A6$ with A , then 4 with $A \Rightarrow 6$

$\$0a2b4b4\cancel{A6} \rightarrow \$0a2b4A6$
 $\quad \quad \quad A \rightarrow bA \quad \quad \quad \downarrow \nearrow$
 $\quad \quad \quad \quad \quad \quad \quad 6$

$w_2 = baba$

$w_3 = ab/c/c$

α	β	Π
\$0	baba\$	ϵ
\$0err	baba\$	ϵ
error because there is no move in LR(0) for row 0 column b		
\$0	ab/c/c\$	ϵ
\$0a2	b/c/c\$	ϵ
\$0a2b4	c/c\$	ϵ
\$0a2b4c5	c\$	
← reduce		
\$0a2b4A6	c\$	3
← reduce		
\$0a2A3	c\$	2, 3
← reduce		
\$0S1	c\$	2, 3
\$ accept	c\$	2, 3

↓
 $w_2 \notin G$

because is not empty and we reached
accept it means the word is not accepted
 $w_3 \notin G$

Ex 2

$$S' \rightarrow S$$

$$S \rightarrow AA \quad (1)$$

$$A \rightarrow aA^{(2)} \mid b^{(3)}$$

Canonical collection

$$r_0 = \text{closure}(\{[S' \rightarrow S]\}) = \{[S' \rightarrow \cdot S], [S \rightarrow \cdot AA], [A \rightarrow \cdot aA], [A \rightarrow \cdot b]\}$$

$$r_1 = \text{goto}(r_0, S) = \text{closure}(\{[S' \rightarrow S \cdot]\}) = \{[S' \rightarrow S \cdot]\}$$

$$r_2 = \text{goto}(r_0, A) = \text{closure}(\{[S \rightarrow A \cdot A]\}) = \{[S \rightarrow A \cdot A], [A \rightarrow \cdot aA], [A \rightarrow \cdot b]\}$$

$$r_3 = \text{goto}(r_0, a) = \text{closure}(\{[A \rightarrow a \cdot A]\}) = \{[A \rightarrow a \cdot A], [A \rightarrow \cdot aA], [A \rightarrow \cdot b]\}$$

$$r_4 = \text{goto}(r_0, b) = \text{closure}(\{[A \rightarrow b \cdot]\}) = \{[A \rightarrow b \cdot]\}$$

$$\text{goto}(r_1, X) = \emptyset, \forall X \in N \cup \Sigma$$

$$\text{goto}(r_2, A) = \text{closure}(\{[S \rightarrow AA \cdot]\}) = \{[S \rightarrow AA \cdot]\} = r_5$$

$$\text{goto}(r_2, a) = \text{closure}(\{[A \rightarrow a \cdot A]\}) = \{[A \rightarrow a \cdot A], [A \rightarrow \cdot aA], [A \rightarrow \cdot b]\} = r_3$$

$$\text{goto}(r_2, b) = \text{closure}(\{[A \rightarrow b \cdot]\}) = r_4$$

$$\text{goto}(r_3, A) = \text{closure}(\{[A \rightarrow aA \cdot]\}) = \{[A \rightarrow aA \cdot]\} = r_6$$

$$\text{goto}(r_3, a) = \text{closure}(\{[A \rightarrow a \cdot A]\}) = r_3$$

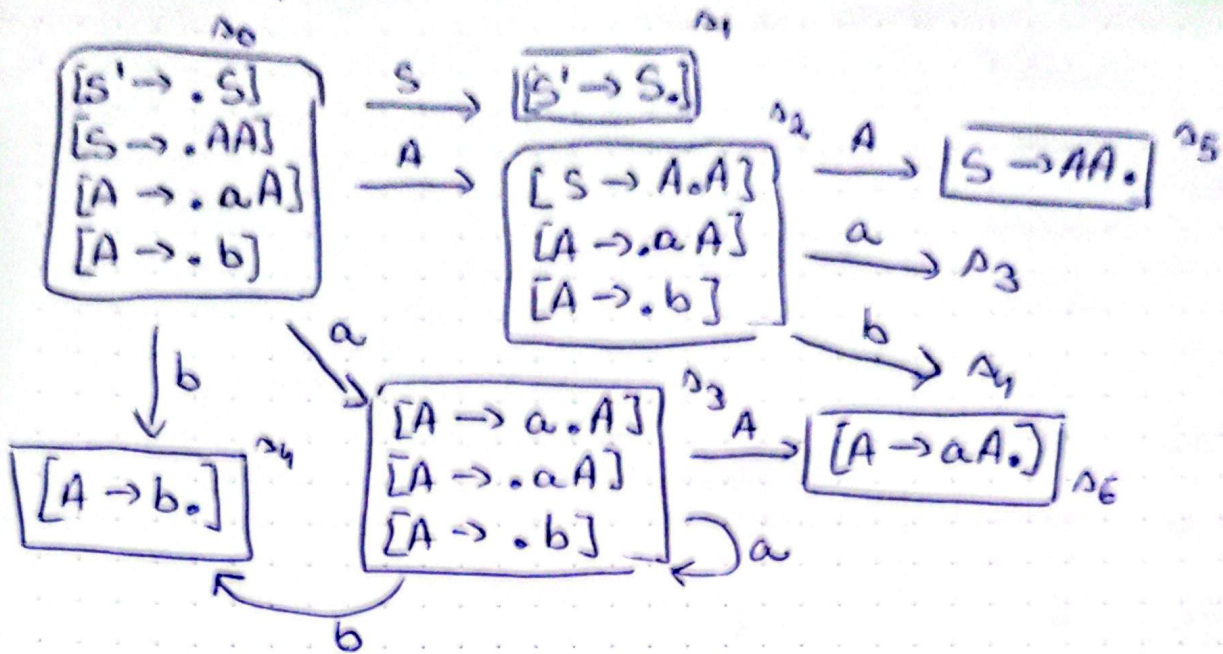
$$\text{goto}(r_3, b) = \text{closure}(\{[A \rightarrow b \cdot]\}) = r_4$$

$$\text{goto}(r_4, X) = \emptyset, \forall X \in N \cup \Sigma$$

$$\text{goto}(r_5, X) = \emptyset, \forall X \in N \cup \Sigma$$

$$\text{goto}(r_6, X) = \emptyset, \forall X \in N \cup \Sigma$$

Visual representation



LR(0) table

State	Action	goto			
		a	b	S	A
0	shift	3	4	1	2
1	accept				
2	shift	3	4		5
3	shift	3	4		6
4	reduce 3				
5	reduce 1				
6	reduce 2				

No conflicts in the table $\Rightarrow G \in LR(0)$

$w_1 = baab$

$w_2 = abba$

$w_3 = babb$

α	β	π
\$0	baab\$	ϵ
\$0b4	aab\$	ϵ
\$0A2	aab\$	3
\$0A2a3	ab\$	3
\$0A2a3a3	b\$	3
\$0A2a3a3b4	\$	3
\$0A2a3a3A6	\$	3,3
\$0A2a3A6	\$	2,3,3
\$0A2A5	\$	2,2,3,3
\$0S1	\$	1,2,3,3
\$accept	\$	1,2,3,3

$\Rightarrow w_1 \in G$

α	β	π
\$0	abba\$	ϵ
\$0a3	bba\$	ϵ
\$0a3b4	ba\$	ϵ
\$0a3A6	ba\$	3
\$0A2	ba\$	2,3
\$0A2b4	a\$	2,3
\$0A2A5	a\$	3,2,3
\$0S1	a\$	3,2,3
\$accept	a\$	3,2,3

$\uparrow$ not empty $\Rightarrow w_2 \notin G$

Ex 2

$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow AA \quad (1) \\ A &\rightarrow aA^2 \mid b^3 \end{aligned}$$

Ex 3

Are the following grammars LR(0)?

a) $S' \rightarrow S$

$S \rightarrow aSbS \quad (1)$

$S \rightarrow a^2 \quad (2)$

b)

$S' \rightarrow S$

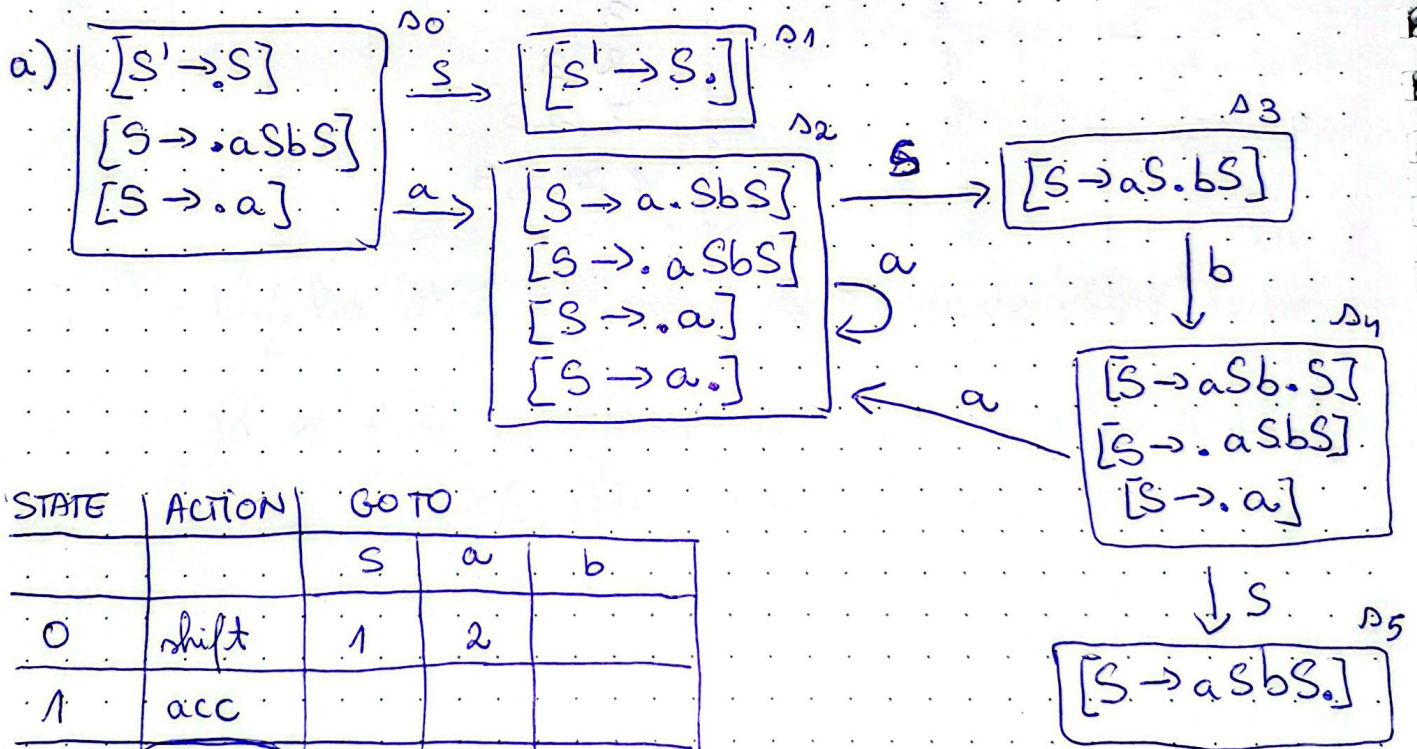
$S \rightarrow Ax \quad (1)$

$S \rightarrow By \quad (2)$

$A \rightarrow a \quad (3)$

$B \rightarrow a \quad (4)$

HW



STATE	ACTION	GOTO		
		S	a	b
0	shift	1	2	
1	acc			
2	reduce & shift			

conflict shift-reduce $\Rightarrow$ grammar $\notin$ LR(0)